EERoom

Browser-Based Electrical Engineering Simulation Platform

Milestone 5: Coding and Testing (3rd Code Iteration)

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Live Application	https://www.eeroom.io
Source Repository	https://github.com/AidenVargas03/EERoom

Application Demonstration (Screencast)	**https://www.loom.com/share/2ef2ed18695b4c9fb6d547853ee5c213 Also listed in Section 9**

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Abstract

EERoom is a free, browser-based electrical engineering simulation platform with an Ohm's Law and power calculator, a waveform visualizer, and a logic gate sandbox. This document records the third code iteration. It completed the two items deferred at Milestone 4: a full password reset flow, and guest access so visitors can use every tool before creating an account, with their work saved to the new account when they log in. It also made every page usable on a tablet, raised text contrast to the WCAG 2.2 AA level, and added an automated test suite for the server API. The project now has 119 automated tests: 82 for the frontend logic and 37 that send real HTTP requests through the Express API. All of them pass. The document contains the Implementation Plan, the Traceability Matrix, the Module Test Cases, the Unit and Integration Test Plans with the Test Analysis Approval Form, and the Source Code Listing required by the CST-452 Capstone Project Handbook.

History and Signoff Sheet

Change Record

Date	Author	Revision Notes
10/11/2026	Aiden Vargas	1.0 Milestone 4 submission: second code iteration, test documentation, and scope adjustments
10/25/2026	Aiden Vargas	2.0 Milestone 5 submission: third code iteration (password reset, guest access, tablet and contrast fixes, API test suite)

Overall Instructor Feedback and Comments

Integrated instructor feedback into project documentation	☐ Yes ☐ No
Scope adjustments approved (Section 2)	☐ Mentor ☐ Professor Thompson
Milestone approval	☐ Professor Thompson

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Appendix A: References

1. Iteration Summary

The second iteration delivered every tool, the dashboard, sharing, and the production deployment. This third iteration closed the gaps that iteration left open and added the automated server testing that Milestone 4 listed as a known limitation.

1.1 Delivered in This Iteration

- Password reset (completes FR-04). A Forgot password? link on the login page opens a form that requests a reset email. The link in the email opens a new Choose a new password page, and the server checks the recovery token with Supabase before it changes anything. The request always returns the same message, so the form cannot be used to find out which emails are registered, and the token is removed from the address bar as soon as it is read.
- Guest access to the tools (resolves SC-06). Visitors can open and use all three tools from the home page without an account. Where the Save button would be, a guest sees Sign up to save and Log in to save. Their work is held in session storage while they register or log in, and it is saved to the account and reopened automatically afterward. The dashboard still requires a login, and the API still rejects every save without a valid token.
- Tablet layout and accessibility (BL-18). Every page was checked at 768 px and 1024 px wide. The logic board, which was fixed at 812 px and partly hidden on a tablet, now scales to fit, and its pin hit areas were enlarged so they stay at least 24 px across, the WCAG 2.2 target size. Three text colors that measured below the WCAG AA contrast minimum of 4.5:1 were lightened until they pass.
- Automated API tests (BL-16). 37 tests send real HTTP requests through the Express app, routes, JWT middleware, and controllers with Supertest. Only Supabase is replaced, by an in-memory stand-in, so the tests run in about two seconds without touching production data. They cover registration and its rollback, login, password reset, the rate limiter, every project route including ownership checks, and public share links.
- Fourteen new frontend unit tests for the reset link parser and the guest draft storage, bringing the frontend total from 68 to 82.

1.2 Technology Stack

Layer	Technology	Hosting
Frontend	React 18, React Router 6, Vite 5, Tailwind CSS 3, Axios	Vercel (www.eeroom.io)
Backend API	Node.js, Express 4, jsonwebtoken, express-rate-limit	Railway
Database and Auth	PostgreSQL and Supabase Auth (bcrypt hashing, password recovery email)	Supabase
Testing	Vitest 1.6 (frontend unit tests and server API tests), Supertest 7, manual system test scripts	Local and production
Version Control	Git and GitHub	github.com/AidenVargas03/EERoom

2. Scope Status

Milestone 4 recorded seven scope adjustments for mentor and instructor approval. Their status after this iteration is below, followed by one new tool change made in this iteration.

ID	Change	Status at Milestone 5
SC-01	Removed the DC Circuit Simulator (FR-06 to FR-11)	Unchanged. Submitted for approval at Milestone 4
SC-02	Removed Google OAuth login (FR-02)	Unchanged. Submitted for approval at Milestone 4
SC-03	Removed Stripe payments, the Pro tier, the save limit, and PDF export (FR-32 to FR-36)	Unchanged. Submitted for approval at Milestone 4
SC-04	Click-to-place grid in SVG instead of react-konva drag and drop (FR-17, FR-18)	Unchanged. The grid now also scales to fit tablet screens
SC-05	Ideal-versus-sampled overlay instead of a three-signal overlay (FR-25)	Unchanged. Submitted for approval at Milestone 4
SC-06	Tools required a login; guest access deferred	Resolved in this iteration (BL-17). Guests can use every tool
SC-07	Password reset form deferred (FR-04)	Resolved in this iteration (BL-14). FR-04 is now fully implemented

2.1 Tool Change in This Iteration

ID	Change	Justification
SC-08	Server API tests use Vitest with Supertest instead of the Jest with Supertest named in the Milestone 4 backlog (BL-16)	Vitest already runs the frontend tests, so one test runner and one set of commands now covers both halves of the project. It supports the server's ES modules without extra configuration. Supertest, the part that sends the HTTP requests, is unchanged.

3. Implementation Plan

Delivery continued with Kanban. This iteration carried the four items planned for Milestone 5 in the backlog (BL-14, BL-16, BL-17, BL-18) and one item for the milestone deliverables (BL-20). Estimates come from the Milestone 4 backlog. Actual hours are the developer's own record, about 8 hours in total for the iteration.

Use Case or User Story	List of Detailed Development Tasks	Estimate (hrs.)	Actual (hrs.)	Percent Complete
BL-14 As a student, I want to reset a forgotten password.	Forgot Password page; POST /auth/reset-password sends a fixed redirect URL and returns the same message for every email; POST /auth/update-password verifies the recovery token with Supabase before changing the password; Reset Password page; resetLink.js parser with 7 unit tests; PASSWORD_RESET_URL on Railway and redirect URLs in Supabase	3	2	100%
BL-16 As the developer, I want automated integration tests for the server API.	Split app.js out of index.js so tests can import the app; in-memory Supabase stand-in; 37 Supertest tests across auth, password reset, rate limiting, projects, and sharing; checked by temporarily removing an ownership filter and confirming a test failed	6	2	100%
BL-17 As a visitor, I want to try the tools before I register.	Public tool routes; GuestSavePrompt in place of Save for guests; pendingDraft.js session storage with 7 unit tests; LoginPage saves the draft after login and opens it; Back to Home for guests; Try it now links on the landing page	3	1.5	100%
BL-18 As a student on a tablet, I want every page to work at 768 px and above.	Checked all 10 pages at 768 px and 1024 px; logic board scales to fit; pin hit area kept at 24 px or more; three text colors raised to WCAG AA contrast; keyboard tab order checked on the forms	4	1	100%
BL-20 As the developer, I want Milestone 5 documentation, a manual test run, and a screencast.	This document; Requirements Specification Rev 1.2; backlog and traceability workbooks; manual test cases MT-18 to MT-27 and a regression run of MT-01 to MT-17; screencast	6	1.5	100%
		Total estimate	**Total actual**	
		22	8	

Percent of user stories complete for this iteration	5 of 5 user stories complete (100%)
Percent of user stories complete for entire project	19 of 20 backlog items complete (95%). BL-20 was added for this milestone's deliverables. Remaining for Milestone 6: BL-19 final presentation and release.

Actual hours came in under the estimates because the backend reset endpoint, the project API, and the test-friendly structure of the evaluators already existed from earlier iterations, so most items needed new pages and tests rather than new infrastructure.

4. Functional Requirements Mapping (Traceability Matrix)

Each requirement is mapped to the Milestone 3 Architecture Plan section where it was designed, the file and function that implement it, and the test cases that verify it. UT cases are frontend unit tests, IT cases are automated server API tests, and MT cases are manual system tests. Rows changed in this iteration are marked (Iteration 3).

Requirement (Use Case or User Story)	Architecture Plan (Milestone 3 Section)	Code Module (Class and Method)	Test Case	Status
FR-01 Register with email and password	4.6 API, 4.7.1 Registration Sequence	authController.register() RegisterPage.jsx, useAuth.register()	MT-01, MT-02 IT-01 to IT-04	Implemented
FR-02 Log in with Google OAuth	N/A	N/A	N/A	Removed (SC-02)
FR-03 Log out from any page	4.9.2 Dashboard	useAuth.logout() DashboardPage.jsx	MT-05	Implemented
FR-04 Reset password by email (Iteration 3)	4.6 API	authController.resetPassword(), updatePassword() ForgotPasswordPage.jsx, ResetPasswordPage.jsx resetLink.parseResetLink()	MT-18 to MT-21 IT-09 to IT-15 UT-17	Implemented
FR-05 Error message for invalid login	4.6 API	authController.login() LoginPage.jsx	MT-03 IT-06 to IT-08	Implemented
FR-06 to FR-11 Circuit simulator	N/A	N/A	N/A	Removed (SC-01)
FR-12, FR-13 Enter two of V, I, R and calculate the third	4.8.1 Ohm's Law Algorithm, 4.9.3	OhmToolPage.jsx (solve-for logic)	MT-06	Implemented
FR-14 Calculate power	4.8.1	OhmToolPage.jsx (P = V x I)	MT-06	Implemented
FR-15, FR-16 Instant updates, units, and invalid input handling	4.9.3	OhmToolPage.jsx (zero-division guards)	MT-06, MT-07	Implemented
FR-17 Logic gate components	4.3.3 LogicGateSandbox, 4.9.4	logicEvaluator.GATES, inputCountFor() LogicToolPage.jsx palette	UT-01, UT-02	Implemented (SC-04)
FR-18 Connect gates with wires	4.7.3 Logic Sequence	LogicToolPage.jsx (pin wiring, boardPointFromEvent()) logicEvaluator.validateCircuit()	UT-06, MT-16, MT-26	Implemented (SC-04)
FR-19 Evaluate gate outputs	4.8.2 Boolean Evaluation	logicEvaluator.evaluateCircuit()	UT-01, UT-03, UT-04	Implemented
FR-20 Generate the truth table	4.8.2	logicEvaluator.generateTruthTable(), toExpression()	UT-05, UT-07, UT-08	Implemented

Requirement (Use Case or User Story)	Architecture Plan (Milestone 3 Section)	Code Module (Class and Method)	Test Case	Status
FR-21 Toggle inputs between 0 and 1	4.9.4	LogicToolPage.jsx (INPUT click handler)	MT-16	Implemented
FR-22 Select a signal type	4.3.3 WaveformVisualizer	waveform.WAVEFORM_TYPES, sampleAt()	UT-09	Implemented
FR-23 Set amplitude, frequency, and phase	4.3.3	waveform.sampleAt() WaveToolPage.jsx controls	UT-10, UT-11, UT-12	Implemented
FR-24 Render an oscilloscope-style display	4.9.5	WaveformCanvas.drawWaveform() waveform.generateSamples()	UT-13, UT-14, MT-15	Implemented
FR-25 Overlay signals for comparison	4.9.5	waveform.isAliased() WaveformCanvas.jsx (ideal and sampled traces)	UT-15, UT-16, MT-15	Changed (SC-05)
FR-26 Label axes with time and voltage	4.9.5	WaveformCanvas.formatTime(), drawWaveform()	MT-15	Implemented
FR-27 Save a project	4.6 API, 4.7.2 Save Sequence	projectController.createProject(), updateProject() api/projects.js	MT-08 IT-21 to IT-24, IT-28	Implemented
FR-28 Load projects from the dashboard	4.6, 4.9.2	projectController.getAllProjects(), getProjectById() DashboardPage.jsx	MT-09 IT-25, IT-26	Implemented
FR-29 Delete a project	4.6, 4.9.2	projectController.deleteProject() DashboardPage.jsx	MT-10 IT-30	Implemented
FR-30 Generate a shareable URL	4.6 API	projectController.generateShareLink() ShareButton.jsx	MT-11 IT-32, IT-33	Implemented
FR-31 View a shared project without logging in	4.6, 4.10 Security	shareRoutes.js SharedProjectPage.jsx	MT-12, MT-13 IT-35 to IT-37	Implemented
FR-32 to FR-36 Free-tier limit and subscriptions	N/A	N/A	N/A	Removed (SC-03)
SC-06 Use the tools as a guest and keep the work on sign-up (Iteration 3)	4.9.1 Landing Page	App.jsx (public tool routes) GuestSavePrompt.jsx, pendingDraft.js LoginPage.handleSubmit()	MT-22 to MT-25 UT-18	Implemented
NFR-04 to NFR-06 Password hashing, HTTPS, 24-hour tokens	4.10 Security Design	Supabase Auth (bcrypt) authMiddleware.requireAuth() authController.login()	MT-04 IT-05, IT-18 to IT-20	Implemented

Requirement (Use Case or User Story)	Architecture Plan (Milestone 3 Section)	Code Module (Class and Method)	Test Case	Status
NFR-08, NFR-09 Usable on tablets, readable contrast (Iteration 3)	4.9 User Interface	LogicToolPage.jsx (scaling board, PIN_HIT_R) theme.js (contrast values)	MT-26, MT-27	Implemented
NFR-13 Limit login attempts	4.10 Security Design	authRoutes.js (authLimiter)	MT-14 IT-16, IT-17	Implemented
NFR-14 Automated tests for core logic	4.10, Test Plan	client/src/utils/*.test.js server/tests/*.test.js	UT-01 to UT-18 IT-01 to IT-37	Implemented
Security Users can only access their own projects	4.10 Security Design	projectController.js (.eq on user_id)	MT-17 IT-27, IT-29, IT-31, IT-34	Implemented

5. Module Test Cases

Test cases follow the handbook template. UT-01 to UT-18 are automated frontend unit tests and IT-01 to IT-37 are automated server API tests; each step is one automated test, and the results come from the run on October 25, 2026, in which all 119 tests passed. MT-01 to MT-17 are the Milestone 4 manual cases, rerun as a regression check, and MT-18 to MT-27 are new manual cases for this iteration. Manual cases were run by the developer against the production site.

5.1 Frontend Unit Test Cases (Automated)

UT-01: Gate definitions match textbook truth tables

Priority	High
Module	client/src/utils/logicEvaluator.js (GATES)
Test Objective	Validate that each of the seven gate types produces its standard truth table. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	AND produces 0001	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass
2	OR produces 0111	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass
3	NAND produces 1110	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass
4	NOR produces 1000	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass
5	XOR produces 0110	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass
6	XNOR produces 1001	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass

7	NOT produces 10	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass
8	Offers exactly seven gate types	Run npm test in the client folder	Every input combination for the gate	Output bits match the textbook table	Pass

UT-02: Pin counts per component

Priority	Medium
Module	logicEvaluator.js (inputCountFor)
Test Objective	Validate that each component exposes the correct number of input pins. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Gives NOT one pin and the binary gates two	Run npm test in the client folder	Each component type	NOT has 1 pin, binary gates 2, INPUT 0, OUTPUT 1	Pass
2	Gives INPUT no pins and OUTPUT one	Run npm test in the client folder	Each component type	NOT has 1 pin, binary gates 2, INPUT 0, OUTPUT 1	Pass

UT-03: Unconnected pins read as 0 and are reported

Priority	High
Module	logicEvaluator.js (evaluateCircuit)
Test Objective	Validate the error-condition behavior of a gate with a floating input. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Gives AND a 0 output when its second pin is floating and A is high	Run npm test in the client folder	Circuit with one unwired pin	Floating pin evaluates as 0 and is named in the report	Pass
2	Gives OR a 1 output in the same situation	Run npm test in the client folder	Circuit with one unwired pin	Floating pin evaluates as 0 and is named in the report	Pass
3	Names the floating pin so the interface can flag it	Run npm test in the client folder	Circuit with one unwired pin	Floating pin evaluates as 0 and is named in the report	Pass

4	Reports nothing floating once every pin is wired	Run npm test in the client folder	Circuit with one unwired pin	Floating pin evaluates as 0 and is named in the report	Pass

UT-04: Feedback loops are refused

Priority	High
Module	logicEvaluator.js (evaluateCircuit, generateTruthTable)
Test Objective	Validate that a circuit with a loop raises an error instead of returning a guessed value. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Throws when evaluating a loop	Run npm test in the client folder	Circuit with a cycle	An error is thrown	Pass
2	Throws when building a truth table from a loop	Run npm test in the client folder	Circuit with a cycle	An error is thrown	Pass
3	Throws on a gate wired directly back to itself	Run npm test in the client folder	Circuit with a cycle	An error is thrown	Pass

UT-05: Truth table size limits

Priority	Medium
Module	logicEvaluator.js (generateTruthTable)
Test Objective	Validate the ten-input limit and the row ordering of the truth table. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Allows 10 inputs	Run npm test in the client folder	Circuits with 0, 10, and 11 inputs	Within the limit a table is built; above it an error is raised	Pass
2	Refuses one more than the limit	Run npm test in the client folder	Circuits with 0, 10, and 11 inputs	Within the limit a table is built; above it an error is raised	Pass
3	Still gives one row for a circuit with no inputs	Run npm test in the client folder	Circuits with 0, 10, and 11 inputs	Within the limit a table is built; above it an error is raised	Pass
4	Counts rows up in binary with the first input as the most significant bit	Run npm test in the client folder	Circuits with 0, 10, and 11 inputs	Within the limit a table is built; above it an error is raised	Pass

UT-06: Structural validation errors

Priority	High
Module	logicEvaluator.js (validateCircuit)
Test Objective	Validate that malformed circuits are detected and reported. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Catches a duplicate node id	Run npm test in the client folder	Malformed circuit objects	Each structural problem is reported; a valid circuit reports none	Pass
2	Catches an unknown gate type	Run npm test in the client folder	Malformed circuit objects	Each structural problem is reported; a valid circuit reports none	Pass
3	Catches a wire landing on a pin the gate does not have	Run npm test in the client folder	Malformed circuit objects	Each structural problem is reported; a valid circuit reports none	Pass
4	Catches two wires driving one pin, which is a short rather than a circuit	Run npm test in the client folder	Malformed circuit objects	Each structural problem is reported; a valid circuit reports none	Pass
5	Catches a wire into an INPUT component	Run npm test in the client folder	Malformed circuit objects	Each structural problem is reported; a valid circuit reports none	Pass
6	Reports nothing for a well-formed circuit	Run npm test in the client folder	Malformed circuit objects	Each structural problem is reported; a valid circuit reports none	Pass

UT-07: Column ordering and half adder

Priority	Medium
Module	logicEvaluator.js (getInputNodes, getOutputNodes)
Test Objective	Validate column order and correctness against a half adder computed by hand. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Follows the board, so Sum placed above Carry gives columns S then C	Run npm test in the client folder	Half adder circuit (XOR and AND)	Columns follow board order; results match the hand-computed table	Pass
2	Falls back to label order when nodes have no board position	Run npm test in the client folder	Half adder circuit (XOR and AND)	Columns follow board order; results match the hand-computed table	Pass
3	Separates inputs from outputs	Run npm test in the client folder	Half adder circuit (XOR and AND)	Columns follow board order; results match the hand-computed table	Pass

4	Matches the truth table computed by hand	Run npm test in the client folder	Half adder circuit (XOR and AND)	Columns follow board order; results match the hand-computed table	Pass
5	Derives expressions that agree with the gates used	Run npm test in the client folder	Half adder circuit (XOR and AND)	Columns follow board order; results match the hand-computed table	Pass
6	Carries only when both inputs are high	Run npm test in the client folder	Half adder circuit (XOR and AND)	Columns follow board order; results match the hand-computed table	Pass

UT-08: Boolean expression generation

Priority	Medium
Module	logicEvaluator.js (toExpression)
Test Objective	Validate that the derived expression matches how the circuit is wired and evaluated. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Nests sub-expressions in the order the gates are wired	Run npm test in the client folder	Nested gate circuits	Expression nests in wiring order and writes a floating pin as 0	Pass
2	Writes an unconnected pin as 0, matching how it is evaluated	Run npm test in the client folder	Nested gate circuits	Expression nests in wiring order and writes a floating pin as 0	Pass

UT-09: Waveform shape definitions

Priority	High
Module	client/src/utils/waveform.js (sampleAt)
Test Objective	Validate the four waveform shapes at key points in the cycle. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Starts every shape in phase at t=0, except square which has no rising zero crossing	Run npm test in the client folder	t at quarter and half cycles	Each shape produces its expected values	Pass
2	Puts a 1 Hz sine through 0, +1, 0, -1 at the quarter points	Run npm test in the client folder	t at quarter and half cycles	Each shape produces its expected values	Pass
3	Peaks the triangle at the quarter points and crosses zero at the half	Run npm test in the client folder	t at quarter and half cycles	Each shape produces its expected values	Pass

4	Flips the square wave at the half cycle	Run npm test in the client folder	t at quarter and half cycles	Each shape produces its expected values	Pass
5	Ramps the sawtooth and resets it at the half cycle	Run npm test in the client folder	t at quarter and half cycles	Each shape produces its expected values	Pass
6	Exposes exactly the four supported shapes	Run npm test in the client folder	t at quarter and half cycles	Each shape produces its expected values	Pass

UT-10: Waveform parameters

Priority	High
Module	waveform.js (sampleAt)
Test Objective	Validate amplitude, offset, and phase handling. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Scales by amplitude	Run npm test in the client folder	Amplitude, offset, phase values	Output scales, shifts, and phase-shifts correctly	Pass
2	Shifts by DC offset	Run npm test in the client folder	Amplitude, offset, phase values	Output scales, shifts, and phase-shifts correctly	Pass
3	Treats 90 degrees of phase as a quarter cycle	Run npm test in the client folder	Amplitude, offset, phase values	Output scales, shifts, and phase-shifts correctly	Pass
4	Wraps negative phase into the cycle rather than leaving it out of range	Run npm test in the client folder	Amplitude, offset, phase values	Output scales, shifts, and phase-shifts correctly	Pass

UT-11: Waveform edge cases

Priority	Medium
Module	waveform.js (sampleAt)
Test Objective	Validate behavior at 0 Hz, zero amplitude, and negative frequency. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Holds a constant at 0 Hz, because the cycle never advances	Run npm test in the client folder	0 Hz, 0 V, negative Hz	Constant, flat, and mirrored outputs	Pass

2	Produces a flat line at zero amplitude	Run npm test in the client folder	0 Hz, 0 V, negative Hz	Constant, flat, and mirrored outputs	Pass
3	Mirrors the wave for a negative frequency	Run npm test in the client folder	0 Hz, 0 V, negative Hz	Constant, flat, and mirrored outputs	Pass

UT-12: Waveform error conditions

Priority	High
Module	waveform.js (sampleAt)
Test Objective	Validate that invalid input is rejected instead of silently coerced. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Rejects an unknown waveform type	Run npm test in the client folder	Unknown type, NaN, Infinity, numeric strings	An error is thrown for each	Pass
2	Rejects a non-finite time	Run npm test in the client folder	Unknown type, NaN, Infinity, numeric strings	An error is thrown for each	Pass
3	Rejects non-finite parameters	Run npm test in the client folder	Unknown type, NaN, Infinity, numeric strings	An error is thrown for each	Pass
4	Rejects a numeric string rather than silently coercing it	Run npm test in the client folder	Unknown type, NaN, Infinity, numeric strings	An error is thrown for each	Pass

UT-13: Sample generation

Priority	High
Module	waveform.js (generateSamples)
Test Objective	Validate sample count, ordering, and handling of invalid durations and rates. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Includes both endpoints, so one second at 1000 Hz is 1001 samples	Run npm test in the client folder	1 s at 1000 Hz; non-positive values	1001 ordered samples; empty for zero duration; error for a bad rate	Pass

2	Returns times in ascending order	Run npm test in the client folder	1 s at 1000 Hz; non-positive values	1001 ordered samples; empty for zero duration; error for a bad rate	Pass
3	Treats a non-positive duration as nothing to draw, not an error	Run npm test in the client folder	1 s at 1000 Hz; non-positive values	1001 ordered samples; empty for zero duration; error for a bad rate	Pass
4	Rejects a non-positive sample rate, which is incoherent rather than empty	Run npm test in the client folder	1 s at 1000 Hz; non-positive values	1001 ordered samples; empty for zero duration; error for a bad rate	Pass

UT-14: RMS and voltage range

Priority	Medium
Module	waveform.js (rms, voltageRange)
Test Objective	Validate the readout calculations. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Gives a sine peak divided by the square root of 2	Run npm test in the client folder	Sine and square samples; empty set	Sine RMS = peak / sqrt(2); square RMS = amplitude; empty set returns 0	Pass
2	Gives a square wave its own amplitude	Run npm test in the client folder	Sine and square samples; empty set	Sine RMS = peak / sqrt(2); square RMS = amplitude; empty set returns 0	Pass
3	Returns 0 for an empty set rather than NaN	Run npm test in the client folder	Sine and square samples; empty set	Sine RMS = peak / sqrt(2); square RMS = amplitude; empty set returns 0	Pass
4	Spans offset plus and minus amplitude	Run npm test in the client folder	Sine and square samples; empty set	Sine RMS = peak / sqrt(2); square RMS = amplitude; empty set returns 0	Pass
5	Returns zeroes for an empty set	Run npm test in the client folder	Sine and square samples; empty set	Sine RMS = peak / sqrt(2); square RMS = amplitude; empty set returns 0	Pass

UT-15: Nyquist limit check

Priority	High
Module	waveform.js (isAliased)
Test Objective	Validate when a sample rate is flagged as aliased. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail

1	Passes a signal sampled well above twice its frequency	Run npm test in the client folder	Rates above, at, and below 2f	Flagged at or below 2f, never for DC	Pass
2	Flags a signal sampled below twice its frequency	Run npm test in the client folder	Rates above, at, and below 2f	Flagged at or below 2f, never for DC	Pass
3	Flags a signal sampled at exactly twice its frequency, since the theorem needs strictly more	Run npm test in the client folder	Rates above, at, and below 2f	Flagged at or below 2f, never for DC	Pass
4	Uses the magnitude of a negative frequency	Run npm test in the client folder	Rates above, at, and below 2f	Flagged at or below 2f, never for DC	Pass
5	Never flags DC	Run npm test in the client folder	Rates above, at, and below 2f	Flagged at or below 2f, never for DC	Pass

UT-16: Aliasing produces a ghost frequency

Priority	Medium
Module	waveform.js (generateSamples)
Test Objective	Validate that undersampling produces the expected alias frequency. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Shows the true frequency when properly sampled	Run npm test in the client folder	f sampled above and below Nyquist	True frequency when sampled correctly; ghost at |f - fs| when undersampled	Pass
2	Shows a ghost at |f - fs| when undersampled	Run npm test in the client folder	f sampled above and below Nyquist	True frequency when sampled correctly; ghost at |f - fs| when undersampled	Pass

UT-17: Password reset link parsing (Iteration 3)

Priority	High
Module	client/src/utils/resetLink.js (parseResetLink)
Test Objective	Validate that only a recovery link carrying a token is accepted, and that expired links give a readable message. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail

1	Returns the token from a valid recovery link	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass
2	Accepts the hash without a leading #	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass
3	Rejects a token that is not a recovery token	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass
4	Rejects a link with no token	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass
5	Rejects an empty or missing hash	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass
6	Reports Supabase's own message for an expired link	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass
7	Falls back to a generic message when the error has no description	Run npm test in the client folder	Valid, wrong-type, empty, and expired link fragments	Token returned only for a recovery link; a clear error otherwise	Pass

UT-18: Guest draft storage (Iteration 3)

Priority	High
Module	client/src/utils/pendingDraft.js (savePendingDraft, takePendingDraft)
Test Objective	Validate that a guest's work survives sign-up exactly once and that bad data is discarded. (Automated, Vitest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Returns a saved draft unchanged	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass

2	Removes the draft once it has been taken	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass
3	Returns null when nothing is stored	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass
4	Rejects an unknown tool type	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass
5	Rejects a draft with no project data	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass
6	Rejects corrupted JSON and clears it	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass
7	Replaces a missing name with an empty string	Run npm test in the client folder	Valid drafts, unknown tool types, missing data, corrupted JSON	Valid draft returned once then removed; invalid drafts return null	Pass

5.2 Server API Integration Test Cases (Automated, Iteration 3)

IT-01 to IT-08: Registration and login API

Priority	High
Module	server/controllers/authController.js (register, login)
Test Objective	Validate registration, the rollback when the users row fails, the 24-hour JWT, and the vague login error. (Automated, Vitest with Supertest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	IT-01 POST /auth/register	Creates the auth user and the matching users row	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass
2	IT-02 POST /auth/register	Rejects a request with no password	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass

3	IT-03 POST /auth/register	Rejects an email that is already registered	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass
4	IT-04 POST /auth/register	Rolls back the auth user when the users row cannot be written	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass
5	IT-05 POST /auth/login	Returns a 24-hour JWT carrying the user's id	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass
6	IT-06 POST /auth/login	Gives the same vague message for a wrong password	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass
7	IT-07 POST /auth/login	Gives the same vague message for an unknown email	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass
8	IT-08 POST /auth/login	Rejects a login with no email	Test accounts created in the in-memory store	Response status and body match the test's assertions	Pass

IT-09 to IT-15: Password reset API

Priority	High
Module	authController.js (resetPassword, updatePassword)
Test Objective	Validate the reset email redirect, the identical answer for unknown emails, and token and password checks. (Automated, Vitest with Supertest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	IT-09 POST /auth/reset-password	Sends the email with the configured reset page as the redirect	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass
2	IT-10 POST /auth/reset-password	Answers an unknown email exactly like a known one	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass
3	IT-11 POST /auth/reset-password	Rejects a request with no email	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass
4	IT-12 POST /auth/update-password	Changes the password for a valid recovery token, and the new one logs in	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass

5	IT-13 POST /auth/update-password	Refuses an invalid or expired token and changes nothing	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass
6	IT-14 POST /auth/update-password	Refuses a password shorter than 8 characters	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass
7	IT-15 POST /auth/update-password	Refuses a request with no token	Recovery tokens issued by the test store	Response status and body match the test's assertions	Pass

IT-16 to IT-17: Authentication rate limiter

Priority	High
Module	server/routes/authRoutes.js (authLimiter)
Test Objective	Validate that the 11th auth request in 15 minutes is blocked and other routes are not. (Automated, Vitest with Supertest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	IT-16 auth rate limiter	Allows 10 login attempts, then blocks the 11th with 429	11 login attempts from one address	Response status and body match the test's assertions	Pass
2	IT-17 auth rate limiter	Does not limit routes outside /auth	11 login attempts from one address	Response status and body match the test's assertions	Pass

IT-18 to IT-34: Projects API and JWT middleware

Priority	High
Module	server/middleware/authMiddleware.js, server/controllers/projectController.js
Test Objective	Validate token checks and every project route, including that no user can read, edit, delete, or share another user's project. (Automated, Vitest with Supertest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	IT-18 JWT middleware on /projects	Rejects a request with no token	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass

2	IT-19 JWT middleware on /projects	Rejects a token signed with the wrong secret	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
3	IT-20 JWT middleware on /projects	Rejects an expired token	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
4	IT-21 POST /projects	Saves a project owned by the caller	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
5	IT-22 POST /projects	Rejects an unknown tool type	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
6	IT-23 POST /projects	Rejects a project with no data	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
7	IT-24 POST /projects	Ignores a user_id in the body and uses the token's user	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
8	IT-25 GET /projects	Lists only the caller's projects, newest first	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
9	IT-26 GET /projects/:id	Loads the caller's own project	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
10	IT-27 GET /projects/:id	Returns 404 for another user's project	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
11	IT-28 PUT /projects/:id	Renames the caller's project and bumps updated_at	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
12	IT-29 PUT /projects/:id	Cannot edit another user's project	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass

13	IT-30 DELETE /projects/:id	Deletes the caller's project	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
14	IT-31 DELETE /projects/:id	Cannot delete another user's project	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
15	IT-32 POST /projects/:id/share	Creates a 64-character share token and marks the project shared	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
16	IT-33 POST /projects/:id/share	Returns the same token when sharing again	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass
17	IT-34 POST /projects/:id/share	Cannot share another user's project	Two users, ALICE and BOB, with projects of their own	Response status and body match the test's assertions	Pass

IT-35 to IT-37: Public share links

Priority	High
Module	server/routes/shareRoutes.js
Test Objective	Validate public read-only access and that private fields are never returned. (Automated, Vitest with Supertest)

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	IT-35 GET /share/:token	Returns a shared project without logging in, and no private fields	Shared and unshared projects	Response status and body match the test's assertions	Pass
2	IT-36 GET /share/:token	Returns 404 for a token that does not exist	Shared and unshared projects	Response status and body match the test's assertions	Pass
3	IT-37 GET /share/:token	Returns 404 when the project is no longer shared	Shared and unshared projects	Response status and body match the test's assertions	Pass

5.3 Manual System Test Cases

MT-01: Register a new account

Priority	High
Module	RegisterPage.jsx, authController.register()
Test Objective	Validate that a new user can register with a valid email and password.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open register page	Go to https://www.eeroom.io and click Sign up	N/A	Registration form is shown	Pass 10/25/26
2	Submit valid details	Enter name, email, and password, then submit	New email, password of 8+ characters	Redirected to the login page with no error	Pass 10/25/26
3	Confirm stored user	Open the users table in Supabase	Same email	One new row exists for the email	Pass 10/25/26

MT-02: Reject a duplicate registration

Priority	High
Module	authController.register()
Test Objective	Validate that registration fails when the email is already registered.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Register a used email	Submit the registration form again with the MT-01 email	Existing email	An error message is shown and no second account is created	Pass 10/25/26

MT-03: Reject an invalid login

Priority	High
Module	LoginPage.jsx, authController.login()
Test Objective	Validate that a wrong password is refused with a clear message.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail

1	Wrong password	Enter the registered email with a wrong password	Valid email, wrong password	Message: Invalid email or password. User stays on the login page	Pass 10/25/26
2	Correct password	Enter the correct password	Valid credentials	Dashboard opens	Pass 10/25/26

MT-04: Protect private pages

Priority	High
Module	ProtectedRoute.jsx, authMiddleware.requireAuth()
Test Objective	Validate that private pages and API routes cannot be used without logging in.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open dashboard logged out	In a private window, go to https://www.eeroom.io/dashboard	No session	Redirected to the login page	Pass 10/25/26
2	Call API without a token	Send GET /projects to the API with no Authorization header	No token	401 Unauthorized	Pass 10/25/26

MT-05: Log out

Priority	Medium
Module	useAuth.logout(), DashboardPage.jsx
Test Objective	Validate that logging out ends the session.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Log out	Click Log out on the dashboard	Logged-in user	Token is cleared and the login page is shown	Pass 10/25/26
2	Try the back button	Press Back to the dashboard	N/A	Redirected to the login page	Pass 10/25/26

MT-06: Calculate with Ohm's Law

Priority	High
Module	OhmToolPage.jsx

Test Objective	Validate that any two values produce the third value and the power instantly.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Solve for current	Enter V and R	V = 12, R = 4	I = 3 A, P = 36 W	Pass 10/25/26
2	Solve for resistance	Clear R, enter V and I	V = 9, I = 3	R = 3 Ω, P = 27 W	Pass 10/25/26
3	Solve for voltage	Clear V, enter I and R	I = 2, R = 5	V = 10 V, P = 20 W	Pass 10/25/26

MT-07: Handle division by zero

Priority	High
Module	OhmToolPage.jsx
Test Objective	Validate the error condition when a zero would cause division by zero.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Zero current	Enter V and a current of 0	V = 5, I = 0	Message: Current cannot be 0 when solving for resistance	Pass 10/25/26
2	Zero resistance	Enter V and a resistance of 0	V = 5, R = 0	Message: Resistance cannot be 0 when solving for current	Pass 10/25/26

MT-08: Save a new project

Priority	High
Module	projectController.createProject(), api/projects.js
Test Objective	Validate that tool state is saved as a named project.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Save	In the Ohm tool, name the project and click Save	Name: Test Ohm 1	Save confirmation; URL now includes ?project=	Pass 10/25/26
2	Save again	Change a value and click Save	New value	Same project is updated; no duplicate on the dashboard	Pass 10/25/26

MT-09: Reopen a project from the dashboard

Priority	High
Module	DashboardPage.jsx, projectController.getProjectById()
Test Objective	Validate that saved projects are listed and reopen with their saved state.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	View list	Open the dashboard	Projects from MT-08 and MT-15	Projects listed newest first with tool type and date	Pass 10/25/26
2	Open	Click Open on a project	Test Ohm 1	Tool opens with the saved values	Pass 10/25/26

MT-10: Delete a project

Priority	Medium
Module	DashboardPage.jsx, projectController.deleteProject()
Test Objective	Validate that deletion asks for confirmation and removes the project.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Cancel delete	Click Delete, then Cancel	Any project	Confirm and Cancel buttons appear; after Cancel the project remains	Pass 10/25/26
2	Confirm delete	Click Delete, then Confirm	Same project	Project is removed from the list and the database	Pass 10/25/26

MT-11: Create a share link

Priority	High
Module	ShareButton.jsx, projectController.generateShareLink()
Test Objective	Validate that a saved project produces a shareable link.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail

1	Share	Open a saved project and click Share	Saved project	A /share/ link is displayed and copied to the clipboard	Pass 10/25/26

MT-12: View a shared project logged out

Priority	High
Module	shareRoutes.js, SharedProjectPage.jsx
Test Objective	Validate read-only access for a visitor with no account.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open link	Paste the MT-11 link into a private window	Share URL	Project renders read only with no login prompt	Pass 10/25/26
2	Repeat for each tool	Share and open an Ohm, waveform, and logic project	Three share URLs	Each tool type renders correctly	Pass 10/25/26

MT-13: Reject an invalid share link

Priority	Medium
Module	shareRoutes.js
Test Objective	Validate the error condition for a token that does not exist.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Bad token	Go to https://www.eeroom.io/share/notarealtoken	Invalid token	Message: This link is not valid. No project data is shown	Pass 10/25/26

MT-14: Rate limit login attempts

Priority	High
Module	authRoutes.js (authLimiter)
Test Objective	Validate that repeated login attempts are blocked.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail

1	Repeat failed logins	Submit a wrong password 11 times within 15 minutes	Valid email, wrong password	The 11th attempt is refused with a too-many-requests message	Pass 10/25/26

MT-15: Visualize a waveform and its aliasing

Priority	High
Module	WaveToolPage.jsx, WaveformCanvas.jsx
Test Objective	Validate drawing, axis labels, readouts, and the aliasing warning.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Draw each shape	Select sine, square, sawtooth, and triangle	5 V, 50 Hz	Each shape draws with labeled time and voltage axes	Pass 10/25/26
2	Healthy sample rate	Set the sample rate	1000 Hz	Sampled trace sits on the ideal curve; no warning	Pass 10/25/26
3	Undersample	Lower the sample rate	60 Hz	Sampled trace departs from the curve and an aliasing warning appears	Pass 10/25/26

MT-16: Build and evaluate a logic circuit

Priority	High
Module	LogicToolPage.jsx
Test Objective	Validate placement, wiring, toggling, the truth table, and floating-pin warnings.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Place components	Place two INPUTs, an AND gate, and an OUTPUT	A, B, AND, Y	Components snap to grid cells	Pass 10/25/26
2	Leave a pin unwired	Wire only A into the AND gate	N/A	The open pin is ringed in amber and a warning reports 1 unconnected input pin	Pass 10/25/26
3	Finish wiring	Wire B and the AND output to Y	N/A	Truth table shows Y = 1 only when A = 1 and B = 1; expression is Y = (A AND B)	Pass 10/25/26

4	Toggle inputs	Click A and B to set both to 1	A = 1, B = 1	Output Y shows 1	Pass 10/25/26
5	Remove a wire	Select a wire and delete it	Any wire	Wire is removed and results update	Pass 10/25/26

MT-17: Block access to another user's project

Priority	High
Module	projectController.js (ownership filter)
Test Objective	Validate that a user cannot open a project owned by another account.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open another user's ID	Log in as user B and open /tools/ohm?project=<user A's project id>	Two test accounts	API returns 404 Project not found; user A's data is not shown	Pass 10/25/26

MT-18: Request a password reset email (Iteration 3)

Priority	High
Module	ForgotPasswordPage.jsx, authController.resetPassword()
Test Objective	Validate that a user can request a reset link from the login page.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open the form	On https://www.eeroom.io/login click Forgot password?	N/A	Reset your password form is shown	Pass 10/25/26
2	Submit email	Enter the account email and click Send Reset Link	Registered email	Message: If an account exists for that email, a reset link is on its way	Pass 10/25/26

MT-19: Set a new password from the email link (Iteration 3)

Priority	High
Module	ResetPasswordPage.jsx, authController.updatePassword()
Test Objective	Validate that the emailed link lets the user choose a new password and log in with it.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open the link	Click the link in the newest reset email	Reset email from MT-18	Choose a new password page opens on www.eeroom.io/reset-password	Pass 10/25/26
2	Save new password	Enter the same new password twice and click Update Password	Password of 8+ characters	Message: Your password has been updated	Pass 10/25/26
3	Log in	Log in with the new password	New password	Dashboard opens	Pass 10/25/26

MT-20: Reject a used or expired reset link (Iteration 3)

Priority	High
Module	resetLink.parseResetLink(), ResetPasswordPage.jsx
Test Objective	Validate the error condition for a link that has already been used.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Reuse the link	Click the MT-19 email link a second time	Used link	An invalid or expired message and a Request a new reset link link are shown; no password form	Pass 10/25/26

MT-21: Reject mismatched passwords (Iteration 3)

Priority	Medium
Module	ResetPasswordPage.jsx
Test Objective	Validate that the two password fields must match.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Mismatch	From a fresh reset link, type two different passwords and submit	password123 / password124	Message: The two passwords do not match. Password is unchanged	Pass 10/25/26

MT-22: Use every tool as a guest (Iteration 3)

Priority	High
Module	App.jsx (public tool routes), LandingPage.jsx

Test Objective	Validate that a visitor can use all three tools without an account.				
Step	**Test Name**	**Test Steps**	**Test Data**	**Expected Results**	**Pass/Fail**
1	Open tools logged out	In a private window, open each tool from the https://www.eeroom.io home page	No session	Ohm's Law, Waveform, and Logic tools open and calculate normally	Pass 10/25/26
2	Check the save area	Look where the Save button normally is	N/A	Sign up to save and Log in to save are shown instead	Pass 10/25/26

MT-23: Guest work is saved after logging in (Iteration 3)

Priority	High				
Module	GuestSavePrompt.jsx, pendingDraft.js, LoginPage.jsx				
Test Objective	Validate that a guest's work is kept through login and saved to the account.				
Step	**Test Name**	**Test Steps**	**Test Data**	**Expected Results**	**Pass/Fail**
1	Build as guest	In a private window, build a circuit or enter values in a tool	Any tool	Results show as normal	Pass 10/25/26
2	Log in to save	Click Log in to save and log in	Existing account	The tool reopens with the same work, saved, with Update Project and Share shown	Pass 10/25/26

MT-24: Guest work is saved after signing up (Iteration 3)

Priority	High				
Module	GuestSavePrompt.jsx, RegisterPage.jsx, LoginPage.jsx				
Test Objective	Validate the same flow for a brand new account.				
Step	**Test Name**	**Test Steps**	**Test Data**	**Expected Results**	**Pass/Fail**
1	Sign up to save	Build something as a guest, click Sign up to save, and register	New email	Redirected to the login page	Pass 10/25/26
2	Log in	Log in with the new account	New account	The tool reopens with the guest's work saved to the new account	Pass 10/25/26

MT-25: Dashboard still requires a login (Iteration 3)

Priority	High
Module	ProtectedRoute.jsx
Test Objective	Validate that opening the tools to guests did not open the dashboard.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Open dashboard logged out	In a private window, go to https://www.eeroom.io/dashboard	No session	Redirected to the login page	Pass 10/25/26

MT-26: Build a circuit on a tablet (Iteration 3)

Priority	Medium
Module	LogicToolPage.jsx (scaling board)
Test Objective	Validate that the whole logic board is usable at tablet width.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Tablet view	Open the Logic Gate Sandbox in Chrome device mode as iPad Mini (768 px)	iPad Mini	All 7 board columns are visible with no scrolling inside the board	Pass 10/25/26
2	Wire to the last column	Place an INPUT in the first column and an OUTPUT in the last, then wire them	INPUT, OUTPUT	Wire connects and the truth table appears	Pass 10/25/26

MT-27: No sideways scrolling at tablet width (Iteration 3)

Priority	Medium
Module	All pages
Test Objective	Validate that every page fits a 768 px screen.

Step	Test Name	Test Steps	Test Data	Expected Results	Pass/Fail
1	Check each page	At 768 px, open the home, login, register, forgot and reset password, dashboard, three tool, and shared project pages	iPad Mini	No page scrolls horizontally and all controls can be reached	Pass 10/25/26

6. Unit and Integration Test Plans

6.1 Test Approach

EERoom is tested in three layers. The calculation engines and the two new helpers (resetLink.js and pendingDraft.js) are pure functions with no React or database code, so they are unit tested in isolation for normal input, edge cases, and error conditions. The server is tested as a whole through its HTTP interface: each API test sends a real request through the Express app, the routes, the JWT middleware, and the controllers, with only Supabase replaced by an in-memory stand-in. Finally, the deployed system is tested end to end with the manual cases in Section 5.3. Defects found at any stage go back to development, are fixed, and are retested.

6.2 Test Tools

#	Tool	Purpose
1	Vitest 1.6	Runs the frontend unit tests and the server API tests
2	Supertest 7	Sends HTTP requests to the Express app inside the server tests without opening a port
3	Google Chrome and DevTools	Manual system testing, device mode for tablet widths, network inspection
4	Supabase dashboard	Verifying rows created, updated, and deleted, and the password reset email settings
5	Browser private window	Testing guest access, logged-out pages, and shared links
6	GitHub	Version history and traceability of fixes

6.3 Component Testing

Component	How It Is Tested	Test Cases
Logic evaluator	Automated unit tests	UT-01 to UT-08 (35 tests)
Waveform engine	Automated unit tests	UT-09 to UT-16 (33 tests)
Reset link parser and guest drafts	Automated unit tests	UT-17, UT-18 (14 tests)
Authentication API	Automated API tests and manual tests	IT-01 to IT-17; MT-01 to MT-05, MT-14, MT-18 to MT-21
Projects and sharing API	Automated API tests and manual tests	IT-18 to IT-37; MT-08 to MT-13, MT-17
Tool pages and guest access	Manual tests of each tool	MT-06, MT-07, MT-15, MT-16, MT-22 to MT-24
Layout and accessibility	Manual tests in Chrome device mode	MT-26, MT-27

6.4 Integration Testing

Integration testing is now automated for the server. The 37 API tests exercise the routes, middleware, and controllers together, which is where the security rules live: a forged or expired

token is refused (IT-19, IT-20), a user ID sent in a request body is ignored in favor of the token's (IT-24), and no route lets one user read, edit, delete, or share another user's project (IT-27, IT-29, IT-31, IT-34). To confirm the tests detect real defects, the ownership filter in getProjectById was removed on purpose; IT-27 failed, and the filter was restored. Frontend-to-backend integration through the browser is covered by the manual flows: register and log in (MT-01, MT-03), save and reopen (MT-08, MT-09), share and view logged out (MT-11, MT-12), and the new guest-to-account flow (MT-23, MT-24).

6.5 Requirements Testing

Every functional requirement in scope maps to at least one test case in the Traceability Matrix (Section 4), and both normal and error conditions are covered. Examples added in this iteration: validate that a password can be reset from the emailed link (MT-19, IT-12); validate that a reset fails for an expired link (MT-20, IT-13); validate that a reset request reveals nothing about which emails exist (IT-10); validate that a guest's work is saved once and only once after login (UT-18, MT-23); and validate that the eleventh login attempt is blocked (IT-16).

6.6 System Test Plan

Part 1: Overview

Date	10/25/2026
Objectives and goals	Confirm that the deployed application meets the in-scope requirements after the third iteration, that the new password reset and guest access flows work end to end, that the site is usable at tablet width, and that nothing from Milestone 4 regressed.
Duration	One test session of about one hour, repeated after any defect fix

Part 2: Scope and Resources

Features to be tested	All Milestone 4 features (regression, MT-01 to MT-17), plus password reset, guest access with save on sign-up, and tablet layout (MT-18 to MT-27)
Features not tested	Removed features (circuit simulator, OAuth, payments). Phone widths below 768 px, which are outside NFR-08
Resources	Developer (Aiden Vargas), two test accounts and one new account for MT-24, production site at www.eeroom.io, Railway API, Supabase project

Part 3: Strategy

Procedures	Run the automated suites (npm test in client and server), then execute MT-01 to MT-27 on the production site, record Pass or Fail and the date, log any defect, fix it, and rerun the affected cases
Environment	Production: Vercel frontend, Railway API, Supabase database, latest Google Chrome on desktop and in device mode
Tools	Vitest, Supertest, Chrome DevTools, Supabase dashboard, private browser window
Pass criteria	All 119 automated tests pass and every High priority manual case passes

Entry Criteria for System Testing

- Unit testing finished: met. 82 of 82 frontend unit tests pass.
- Modules fully integrated: met. 37 of 37 API tests pass, and every page runs against the live API.
- Development complete per the specification: met. All five iteration items are complete.
- Testing environment available: met. The production deployment is used as the staging environment.

6.7 Acceptance Testing

EERoom is a self-directed project with no client, so acceptance is the instructor and mentor approving the system test results through the Test Analysis Approval Form in Section 7. The Technology Capstone Showcase will also give informal feedback from visitors using the live site, which the guest access in this iteration makes possible without an account.

7. Test Analysis Approval Form

Acceptance Test Final Report

Project name	EERoom: Browser-Based Electrical Engineering Simulation Platform
Date	October 25, 2026
General description of the test effort	119 automated tests (82 frontend unit tests and 37 server API tests) and 27 manual system test cases run against the production deployment, including a regression run of the 17 Milestone 4 cases.
Unit test results	82 passed, 0 failed (Vitest 1.6, run on 10/25/2026)
API integration test results	37 passed, 0 failed (Vitest 1.6 with Supertest 7, run on 10/25/2026)
Manual system test results	27 of 27 passed, 0 failed (52 of 52 steps). Date run: 10/25/26

Unresolved Defects and Known Limitations

Issue or Defect	Impact (H, M, L)	Risk Mitigation	Work Around
Supabase's built-in email service sends only a few emails per hour	L	Enough for a class project; a custom SMTP provider can be added in Supabase settings if traffic grows	Wait and request the reset again
API tests replace Supabase with an in-memory stand-in, so database behavior itself is not covered by them	L	Manual cases MT-08 to MT-13 and MT-17 run against the real database	None needed
Layouts below 768 px (phones) were not part of this iteration	L	Outside NFR-08; tools still load on phones but the logic board is small	Use a tablet or computer

Final Sign-Off and Approval

Role	Name	Signature	Date
Developer	Aiden Vargas	Aiden Vargas	10/25/2026
Mentor			
Instructor	Professor Thompson		

8. Source Code Listing

The complete source code is at https://github.com/AidenVargas03/EERoom. Every source file opens with a header comment describing its purpose, and code adapted from official documentation is marked inline with a Source comment and URL. Files added or changed in this iteration are marked (Iteration 3).

8.1 Frontend (client)

File	Lines	Description
src/main.jsx	29	Application entry point. Mounts React with the router and the authentication provider.
src/App.jsx	67	Route table. Tool pages are public; only the dashboard is wrapped in ProtectedRoute. (Iteration 3)
src/theme.js	73	Color palette for Tailwind, the waveform canvas, and the logic board; text colors meet WCAG AA contrast. (Iteration 3)
src/api/client.js	35	Shared Axios instance that sets the API base URL and attaches the JWT to every request.
src/api/projects.js	59	Functions for the /projects endpoints: list, get, create, update, delete, share.
src/hooks/useAuth.jsx	77	React context that stores the logged-in user and token and provides register, login, and logout.
src/components/ProtectedRoute.jsx	36	Redirects to the login page when no user is logged in.
src/components/GuestSavePrompt.jsx	55	Sign up to save and Log in to save buttons shown to guests; stores their work first. (Iteration 3)
src/components/ShareButton.jsx	100	Creates a share link for a saved project, copies it, and displays it.
src/components/WaveformCanvas.jsx	232	Draws the grid, axes, ideal curve, and sampled trace on an HTML5 canvas.
src/pages/LandingPage.jsx	294	Public front page with an aliasing illustration and Try it now links to each tool. (Iteration 3)
src/pages/RegisterPage.jsx	99	Registration form.
src/pages/LoginPage.jsx	126	Login form with a Forgot password? link; saves a guest's pending work after login. (Iteration 3)
src/pages/ForgotPasswordPage.jsx	93	Requests a password reset email. (Iteration 3)
src/pages/ResetPasswordPage.jsx	144	Sets a new password from the emailed link. (Iteration 3)
src/pages/DashboardPage.jsx	270	Lists the user's projects with Open and Delete, quick-launch links, and an empty state.
src/pages/OhmToolPage.jsx	394	Ohm's Law and power calculator with save, reopen, and guest mode. (Iteration 3)

File	Lines	Description
src/pages/WaveToolPage.jsx	372	Waveform Visualizer controls, readouts, aliasing warning, save, reopen, and guest mode. (Iteration 3)
src/pages/LogicToolPage.jsx	894	Logic Gate Sandbox with a board that scales to fit tablets, and guest mode. (Iteration 3)
src/pages/SharedProjectPage.jsx	348	Public read-only view of a shared project for all three tools.
src/utils/logicEvaluator.js	395	Gate definitions, circuit validation, evaluation, truth tables, and boolean expressions.
src/utils/waveform.js	220	Signal generation, sampling, RMS, voltage range, and the Nyquist check.
src/utils/resetLink.js	54	Reads the recovery token or error from a password reset link. (Iteration 3)
src/utils/pendingDraft.js	64	Holds a guest's unsaved work in session storage until they log in. (Iteration 3)
src/utils/logicEvaluator.test.js	381	35 unit tests for the logic evaluator.
src/utils/waveform.test.js	257	33 unit tests for the waveform engine.
src/utils/resetLink.test.js	41	7 unit tests for the reset link parser. (Iteration 3)
src/utils/pendingDraft.test.js	60	7 unit tests for guest draft storage. (Iteration 3)

8.2 Backend (server)

File	Lines	Description
index.js	33	Entry point: loads the environment and starts the server. (Iteration 3)
app.js	90	Builds the Express app: CORS, JSON parsing, proxy trust, routes, and the error handler. Split out so tests can import it. (Iteration 3)
config/supabaseClient.js	72	Two Supabase clients: a service-role client for data and a separate client for sign-in and password recovery.
middleware/authMiddleware.js	53	requireAuth verifies the JWT bearer token and attaches the user to the request.
routes/authRoutes.js	58	Public auth routes, including update-password, with rate limiting of 10 requests per 15 minutes. (Iteration 3)
routes/projectRoutes.js	32	Protected project routes; requireAuth runs on each one.
routes/shareRoutes.js	35	Public GET /share/:token returning only the shared project's content.
controllers/authController.js	217	register, login, resetPassword, and updatePassword. (Iteration 3)
controllers/projectController.js	179	Project create, read, update, delete, and share, each filtered by the owner's user ID.

File	Lines	Description
db/schema.sql	46	PostgreSQL schema for the users and projects tables.
vitest.config.js	12	Test runner settings for the API tests. (Iteration 3)
tests/fakeSupabase.js	232	In-memory stand-in for Supabase used only by the tests. (Iteration 3)
tests/setup.js	22	Test environment variables and the Supabase replacement. (Iteration 3)
tests/helpers.js	16	Creates signed test tokens. (Iteration 3)
tests/auth.test.js	97	IT-01 to IT-08: registration and login. (Iteration 3)
tests/passwordReset.test.js	79	IT-09 to IT-15: password reset. (Iteration 3)
tests/rateLimit.test.js	32	IT-16 to IT-17: auth rate limiter. (Iteration 3)
tests/projects.test.js	171	IT-18 to IT-34: projects API and token checks. (Iteration 3)
tests/share.test.js	40	IT-35 to IT-37: public share links. (Iteration 3)

8.3 Configuration

package.json files define dependencies and scripts for each folder; npm test runs the tests in either folder. vite.config.js, tailwind.config.js, and postcss.config.js configure the frontend build. The server now also needs PASSWORD_RESET_URL set to the reset page's address. Real .env files are excluded from the repository by .gitignore.

9. Application Demonstration

The screencast demonstrates the features completed in this iteration on the live site, www.eeroom.io: the password reset flow from the login page through the email to logging in with the new password; using the tools as a guest and keeping the work after signing up; the logic board at tablet width; and both automated test suites running.

Screencast link	https://www.loom.com/share/2ef2ed18695b4c9fb6d547853ee5c213

Appendix A: References

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